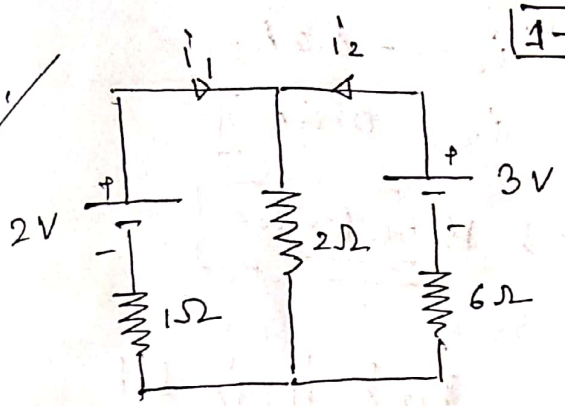
Problems on Wheatstone Bridge, Metre Bridge, Kirchoff's Laws, & Potentiometer.

$\boxed{1-7}$ on Kirchoff's Laws.

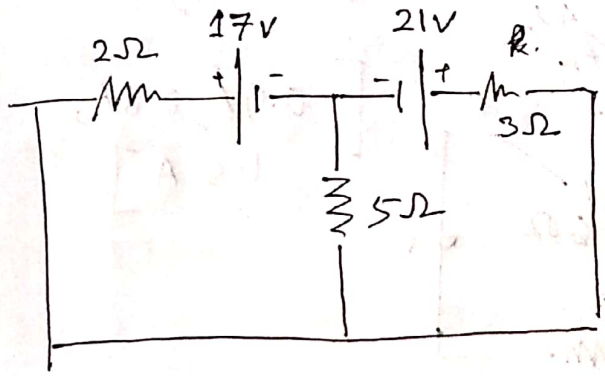
1/



Find i_1 and i_2

[0.5 A, 0.25 A]

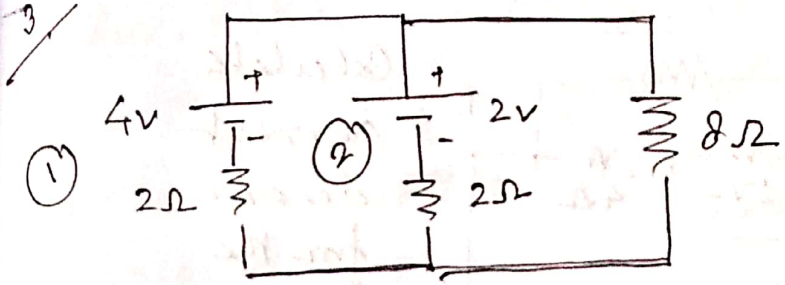
2/



Find currents through 2Ω, 3Ω & 5Ω.

[1 A, 2 A, 3 A]

3/

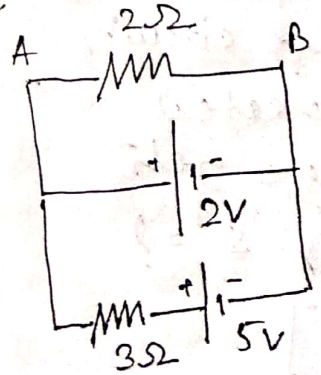


Find magnitude and directions of currents through the cells (1) & (2)

[$\frac{2}{3}$ A from −ve to +ve thro' cell (1)

$\frac{1}{3}$ A from +ve to −ve thro' cell (2)]

4/



Determine current in 3Ω resistance and potential difference betn. A and B in the diagram.

[1 A (right to left), 2 V]

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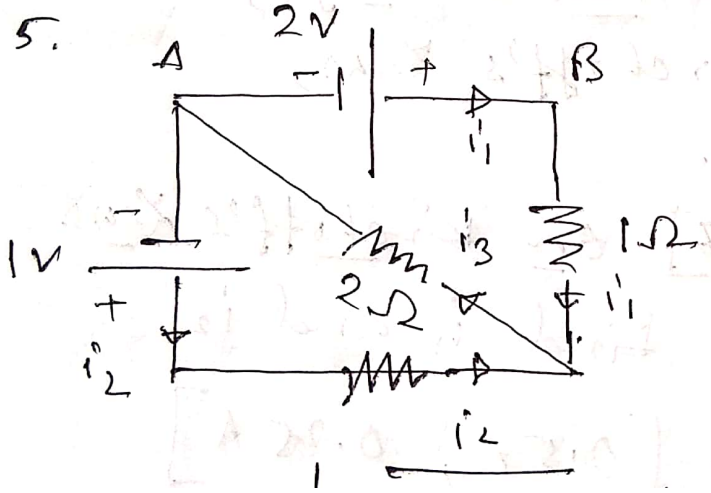
5.



Find i_1, i_2, i_3

[$i_1 = 0.8A$,
$i_2 = -0.2A$,
$i_3 = 0.6A$]

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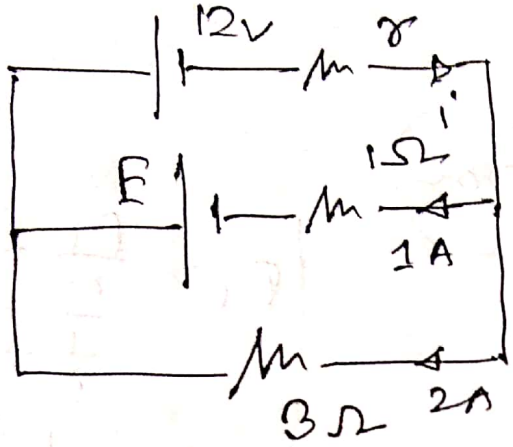
extra problem on kirchoff's KVL.

(#)

6

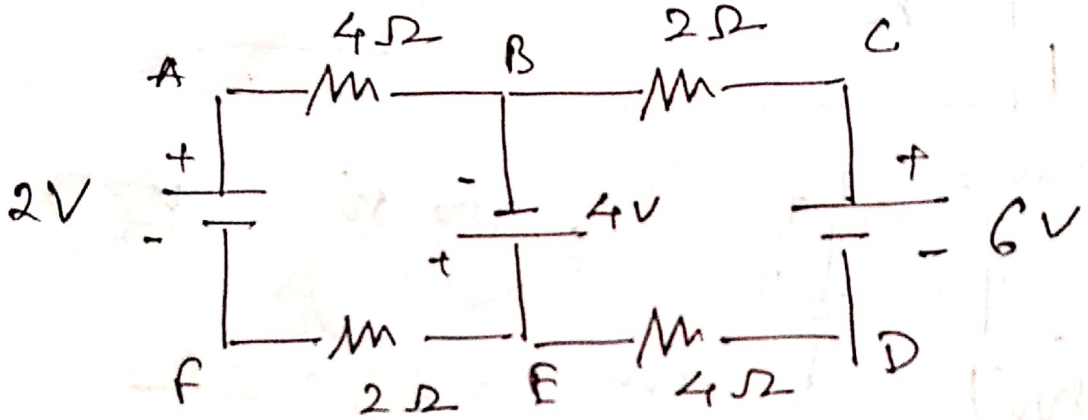


Find r & E

Ans: (2Ω, 5V)

7

(#)



Find current in AB, BE & BC

Ans:

[in AB 1 A (A → B)
in BE $\frac{8}{3}$ A (B → E)
in BC $\frac{5}{3}$ A (C → B)]

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